

Targeting Plasticity Loss in Continual Learning via Null-Space Orthogonal Reorganization

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Abstract. Continual learning (CL) in resource-constrained edge environments is often limited to temporally chunked data access, where historical samples are unavailable and model updates rely solely on locally observed data. While performance degradation is commonly attributed to catastrophic forgetting caused by parameter interference, we show that temporally localized updates can also induce a progressive loss of plasticity, leading to representation collapse and diminished adaptation capability. To address this issue under fixed-capacity and exemplar-free constraints, we propose a diagnostic-and-reorganization framework comprising a multi-dimensional Plasticity Index (PI) and a Null-Space Orthogonal Reorganization (NSOR) mechanism. The PI identifies low-plasticity neurons through activation sparsity, gradient utility, and nonlinear expressivity, whereas NSOR selectively restores their adaptability by constraining updates to directions orthogonal to the dominant historical feature subspace. A transient feature-distillation regularizer is further introduced to stabilize representation evolution during reorganization. Experiments on Split-CIFAR100 and Split-ImageNet100 demonstrate that the proposed approach consistently improves plasticity preservation, reduces forgetting, and achieves competitive performance against representative exemplar-free fixed-capacity continual learning methods.

Keywords: Continual Learning · Plasticity Loss · Neural Tangent Kernel · Orthogonal Reorganization

1 Introduction

Continual Learning (CL) equips artificial intelligence systems with the capability to sequentially learn from non-stationary data streams while preserving previously acquired knowledge [1]. This capability is a prerequisite for open-world embodied intelligence, medical image analysis, and edge computing [2, 3]. In realistic edge scenarios, however, strict physical constraints—limited memory, restricted power, privacy requirements, and intermittent sensing—make it difficult to store or repeatedly revisit historical data. The learner is therefore often restricted to *temporally chunked access*: it can update the model only from the currently visible data block within a limited temporal window [4].

Under fixed-capacity and exemplar-free constraints, performance degradation is traditionally attributed to Catastrophic Forgetting (CF), where new updates

overwrite knowledge acquired from previous tasks [5, 6]. Existing solutions, including regularization, distillation, replay, and network expansion [7], mainly focus on preserving historical knowledge. However, replay buffers, teacher models, or growing architectures are often incompatible with the strict memory and computation budgets of edge devices.

However, preserving old knowledge does not necessarily guarantee the ability to acquire new knowledge. Recent studies further suggest that CF is not the only failure mode in CL. Even when interference is partially controlled, neural networks can gradually lose the ability to learn new information, a phenomenon known as *Loss of Plasticity* (LoP) [8–11]. At the unit level, some neurons may become persistently inactive or gradient-insensitive, which we refer to as low-plasticity or “zombie” units for brevity [12, 13]. At the representation level, this local stagnation is reflected by a decline in the effective rank of the learned manifold. Our controlled analysis indicates that temporally localized updates are associated with such rank reduction even when explicit class-level distribution shift is reduced. This motivates a diagnosis-and-repair perspective: instead of only protecting old parameters, we ask where plasticity is being lost and how to recover underused directions without violating the resource budget. Since plasticity degradation is often accompanied by representation collapse, a representation-level diagnostic is required to monitor the evolution of the learned feature space.

To address this challenge, we propose a plasticity-oriented continual learning framework under fixed-capacity and exemplar-free constraints. The framework combines representation-level diagnosis with targeted neuron reorganization. Specifically, it monitors representation collapse through eNTK effective rank, identifies low-plasticity neurons via a Plasticity Index (PI), and restores their adaptability using Null-Space Orthogonal Reorganization (NSOR) with transient feature distillation.

Unlike prior studies that primarily analyze plasticity degradation as an optimization phenomenon, our work focuses on diagnosing and actively reorganizing low-plasticity neurons under realistic fixed-capacity and exemplar-free constraints. The main contributions of this paper are summarized as follows:

- We reveal that temporally chunked access can inherently accelerate plasticity degradation and representation collapse, even when explicit distribution shift is reduced.
- We propose a multi-dimensional Plasticity Index (PI) to diagnose neuron-level plasticity degradation by jointly modeling activation sparsity, gradient utility, and nonlinear expressivity.
- We introduce an exemplar-free Null-Space Orthogonal Reorganization (NSOR) framework with transient manifold distillation, which restores plasticity while preserving a fixed backbone parameter budget.

2 Related Work

Catastrophic Forgetting Mitigation in Continual Learning. The primary challenge in Continual Learning (CL) is often framed as overcoming Catastrophic Forgetting (CF), i.e., parameter conflict between new and old tasks [14, 15]. Mainstream methods fall into regularization, replay, and dynamic architecture families. Regularization-based methods such as EWC [16], SI [17], and MAS [18], estimate parameter importance and penalize updates on important directions. They are memory efficient, but may over-constrain learning when the task sequence becomes long. Distillation methods such as LwF [19] use a previous model as a teacher, but the output-level constraint alone cannot directly recover inactive or low-utility units. Experience replay (e.g., iCaRL [21]) and generative replay [22, 23] often achieve strong accuracy but require raw samples or generative models. Dynamic expansion methods such as DER [24] reduce interference by adding branches, at the cost of growing capacity. Our work focuses on the stricter case where raw exemplars and backbone growth are both disallowed.

Resource-Constrained Continual Learning. Under edge constraints, recent methods avoid raw-sample replay and model expansion by using parameter isolation, distillation, or prototype statistics. PASS [20] augments prototypes and self-supervised signals to support exemplar-free class-CL, while PlastIL [25] explicitly studies the plastic-stable balance without exemplar buffers. These methods are closer to realistic memory budgets than replay-based approaches, yet they still mainly optimize the external classification objective or protect existing representations. In temporally chunked streams, the feature space itself can become progressively less plastic. This motivates an internal diagnostic that measures which units lose activation, gradient utility, and nonlinear expressivity.

Plasticity Loss and Network Pathology. Recent studies have shifted attention from catastrophic forgetting to the intrinsic learning capability of neural networks [27]. Dohare et al. [8] first demonstrated that deep networks may progressively lose plasticity during long-term non-stationary training, even when CF is partially controlled. Subsequent studies investigated the causes of this phenomenon from the perspectives of optimization dynamics, representation collapse, gradient churn, and feature saturation [9, 13]. These findings suggest that continual learning failure may arise not only from knowledge overwriting but also from the gradual loss of the network’s ability to absorb new information.

Geometric Decoupling for Stability and Plasticity. Orthogonal projection and subspace decoupling have been widely used to reduce interference between old and new knowledge [14]. Existing methods typically constrain updates within carefully designed subspaces to preserve previously learned representations. While effective for stability, such constraints may also reduce adaptation flexibility when model capacity is limited. This observation motivates exploring projection mechanisms that balance representation preservation and plasticity recovery. In summary, existing continual learning research mainly focuses on preserving past knowledge, while progressive plasticity loss under fixed-capacity, exemplar-free settings remains underexplored. This motivates our work on diagnosing and restoring network plasticity.

Null-Space Orthogonal Reorganization (NSOR)

diagnose plasticity loss, then awaken unused capacity in directions that minimally interfere with old features

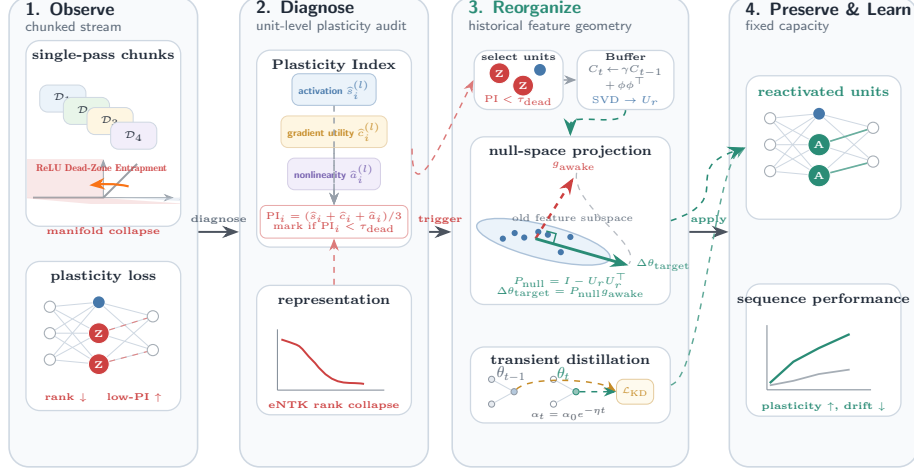


Fig. 1. Algorithmic overview of the proposed framework. (a) Temporally chunked access produces biased local gradients under single-pass, exemplar-free learning. (b) eNTK effective rank and the Plasticity Index jointly diagnose representation collapse and low-plasticity units. (c) Diagnosed units are reactivated by projecting awakening updates into the null space of the historical feature subspace, followed by transient distillation to reduce feature drift. (d) The model keeps a fixed backbone while recovering spare plastic capacity for future chunks.

3 Proposed Method

In temporally chunked continual learning, repeated optimization on localized data may progressively reduce the network’s ability to absorb new information. To address this issue, we propose a plasticity-oriented reorganization framework that consists of two stages: (1) diagnosing low-plasticity neurons through a multi-dimensional Plasticity Index (PI), and (2) selectively restoring their adaptability via Null-Space Orthogonal Reorganization (NSOR). The overall pipeline is illustrated in Fig. 1.

3.1 Problem setting and resource assumptions.

We consider class-incremental learning with a sequence of tasks $\{\mathcal{D}_1, \dots, \mathcal{D}_T\}$. At task t , the learner can access only samples from the current data chunk $\mathcal{D}_t$ and cannot store or replay raw exemplars from previous chunks. The backbone architecture and the number of trainable backbone parameters remain fixed throughout the sequence. Following common exemplar-free protocols, non-raw auxiliary statistics are allowed if their size is independent of the number of

seen images. Our method maintains two bounded auxiliary objects: an exponential moving average of target-layer feature covariance and a temporary teacher snapshot used only during the early iterations after reorganization. Therefore, the method is exemplar-free and fixed in backbone capacity, but it is not strictly zero-memory. We explicitly report this auxiliary state in the experiments.

3.2 Tracing Plasticity Loss via eNTK

Before introducing our intervention mechanism, we quantify the structural degradation associated with temporal chunking. Under strict memory constraints, the model relies solely on the current data chunk for updating. In the controlled pathology study, we additionally measure the angular mismatch between the current-chunk gradient and a reference gradient estimated from the corresponding full training distribution:

$$\cos(\theta_t) = \frac{\langle g_t, g_{\text{ref}} \rangle}{\|g_t\|_2 \|g_{\text{ref}}\|_2} \quad (1)$$

where g_t is the instantaneous gradient computed on the current chunk, and g_{ref} is used only for offline analysis rather than online training. The deployed method never requires access to historical samples.

Persistent gradient mismatch can alter the network’s representation manifold. To monitor this phenomenon without incurring the curse of dimensionality, we employ the empirical Neural Tangent Kernel (eNTK). For a mini-batch of samples X_B , the vectorized Jacobian $J_B(\theta) = \nabla_{\theta} f(X_B; \theta)$ is utilized to construct the eNTK matrix:

$$K_B(\theta) = J_B(\theta) J_B(\theta)^T \quad (2)$$

By applying eigenvalue decomposition to $K_B(\theta)$, we calculate the Effective Rank (ER) of the representation space using the Shannon entropy of its normalized eigenvalue spectrum p_i :

$$ER = \exp \left(- \sum_{i=1}^k p_i \ln p_i \right) \quad (3)$$

A continuous decline in ER indicates that the representation is becoming increasingly low-rank and anisotropic.

3.3 Multi-dimensional Plasticity Index

Plasticity degradation cannot be reliably characterized by a single statistic. For example, a neuron may remain frequently activated while contributing little to optimization, or exhibit non-zero gradients despite being trapped in a low-expressivity regime. We therefore define a multi-dimensional Plasticity Index (PI) that jointly captures activation behavior, optimization utility, and nonlinear expressivity. To identify the micro-level units associated with representation

collapse, we define a normalized Plasticity Index (PI) for each neuron. For the i -th neuron in layer l , the raw statistics are activation frequency $s_i^{(l)}$, gradient utility $c_i^{(l)}$, and nonlinear expressivity $a_i^{(l)}$. Each statistic is tracked by exponential moving average with decay ρ and then normalized within the corresponding layer:

$$\hat{x}_i^{(l)} = \frac{x_i^{(l)} - \min_j x_j^{(l)}}{\max_j x_j^{(l)} - \min_j x_j^{(l)} + \epsilon}, \quad x \in \{s, c, a\}. \quad (4)$$

The activation statistic measures the fraction of non-zero post-activation responses:

$$s_i^{(l)} \leftarrow \rho s_i^{(l)} + (1 - \rho) \frac{1}{|\mathcal{B}|} \sum_{x \in \mathcal{B}} \mathbb{I} [h_i^{(l)}(x) > 0]. \quad (5)$$

The gradient utility estimates whether a neuron contributes to optimization:

$$c_i^{(l)} \leftarrow \rho c_i^{(l)} + (1 - \rho) \left\| \frac{\partial \mathcal{L}}{\partial h_i^{(l)}} h_i^{(l)} \right\|_1. \quad (6)$$

The nonlinear expressivity term measures whether the pre-activation values remain concentrated in the inactive region of the nonlinearity. For ReLU networks, we use the distance between the empirical pre-activation distribution $q_i^{(l)}(z)$ and an inactive reference distribution $q_{\text{dead}}(z)$:

$$a_i^{(l)} = W_1 \left(q_i^{(l)}(z), q_{\text{dead}}(z) \right), \quad q_{\text{dead}}(z) \text{ supported on } z \leq 0. \quad (7)$$

In implementation, $q_i^{(l)}$ is estimated by a fixed-bin histogram of recent pre-activations, and q_{dead} is the empirical distribution obtained by clipping the same activations to the non-positive region. This makes the one-dimensional Wasserstein distance inexpensive to compute and avoids storing raw samples.

The final PI is computed as

$$\text{PI}_i^{(l)} = \lambda_s \hat{s}_i^{(l)} + \lambda_c \hat{c}_i^{(l)} + \lambda_a \hat{a}_i^{(l)}, \quad \lambda_s + \lambda_c + \lambda_a = 1. \quad (8)$$

Unless otherwise stated, we use equal weights $\lambda_s = \lambda_c = \lambda_a = 1/3$. A neuron is marked as low-plasticity when $\text{PI}_i^{(l)} < \tau_{\text{dead}}$. In practice, τ_{dead} is selected on the validation split of the training stream by maximizing the correlation between the low-plasticity ratio and the subsequent drop in eNTK effective rank. The same threshold is then fixed for all test streams.

3.4 Null-Space Orthogonal Reorganization

Instead of constraining all parameter updates as in conventional projection-based continual learning methods, NSOR only modifies updates associated with diagnosed low-plasticity neurons. This targeted design preserves the learning flexibility of healthy neurons while restoring dormant capacity. When the low-plasticity

ratio exceeds a threshold or the current chunk produces a sudden prediction-error spike, the method triggers a targeted reorganization step. The goal is not to reset the whole network, but to update only diagnosed low-plasticity units while reducing first-order perturbations to dominant historical feature directions.

Historical feature subspace. Because raw historical samples are not stored, we maintain a bounded exponential moving average of the uncentered covariance matrix of target-layer features:

$$C_t = \gamma C_{t-1} + (1 - \gamma) \frac{1}{|\mathcal{B}|} \sum_{x \in \mathcal{B}} \phi_\theta(x) \phi_\theta(x)^\top, \quad (9)$$

where $\phi_\theta(x)$ denotes the feature vector at the target layer and γ is the decay factor. We use the penultimate feature layer as the default target layer, so $C_t \in \mathbb{R}^{d \times d}$ has a fixed size independent of the number of observed samples. We compute a truncated singular value decomposition $C_t \approx U_r \Sigma_r U_r^\top$, where r is chosen to retain η of the cumulative spectral energy. The projection matrix onto the orthogonal complement of this principal feature subspace is

$$P_{\text{null}} = I - U_r U_r^\top. \quad (10)$$

Layer-wise projected awakening. For a target linear layer written as $y = W \phi_\theta(x)$, let ΔW_{ideal} be the gradient update for the diagnosed units. We temporarily replace hard inactive responses with a smooth surrogate activation when computing this awakening gradient, which avoids the zero-gradient trap of dead ReLU units. The update is then projected on the feature/input dimension:

$$\Delta W_{\text{target}} = \Delta W_{\text{ideal}} P_{\text{null}}. \quad (11)$$

For convolutional layers, the same operation is applied after flattening the local input patches or, in our default implementation, after the convolutional backbone has produced the penultimate feature vector. This projection reduces first-order perturbations along high-energy historical feature directions, while still allowing updates in underused directions of the representation space.

Transient feature distillation. Because reactivated units can still introduce short-term feature oscillations, we keep a temporary snapshot θ_{old} from the previous chunk and regularize the first iterations after reorganization:

$$\mathcal{L} = \mathcal{L}_{\text{ce}} + \alpha_0 \exp(-\beta k) \|\phi_\theta(x) - \phi_{\theta_{\text{old}}}(x)\|_2^2. \quad (12)$$

The distillation weight decays rapidly, so it stabilizes the transition without turning into a permanent constraint on new-task learning.

Computation and memory. NSOR introduces no extra trainable backbone parameters. The additional state is the covariance matrix C_t , the PI running statistics, and one temporary teacher snapshot. The truncated SVD is executed only when reorganization is triggered, not at every iteration. In our experiments, this trigger-based design keeps the diagnostic overhead small compared with standard backpropagation.

Algorithm 1 Plasticity-Guided Null-Space Reorganization

Require: Current chunk $\mathcal{D}_t$, model f_θ , covariance C_{t-1} , PI threshold τ_{dead} , energy ratio η

- 1: Train f_θ on $\mathcal{D}_t$ with the standard task loss
- 2: Update PI statistics for target layers using mini-batch activations and gradients
- 3: Update feature covariance C_t by exponential moving average
- 4: Compute eNTK effective rank on a held-out mini-batch from the current chunk
- 5: **if** low-plasticity ratio exceeds threshold or prediction error spikes **then**
- 6: Identify neurons with $\text{PI} < \tau_{\text{dead}}$
- 7: Compute truncated SVD $C_t \approx U_r \Sigma_r U_r^\top$
- 8: Construct $P_{\text{null}} = I - U_r U_r^\top$
- 9: Compute layer-wise awakening gradient ΔW_{ideal} for diagnosed neurons
- 10: Apply feature-space projected update $\Delta W_{\text{target}} = \Delta W_{\text{ideal}} P_{\text{null}}$
- 11: Optimize with transient distillation $\mathcal{L} = \mathcal{L}_{\text{ce}} + \alpha_0 \exp(-\beta k) \|\phi_\theta(x) - \phi_{\theta_{\text{old}}}(x)\|_2^2$
- 12: **end if**
- 13: Store only bounded PI statistics, covariance statistics, and a temporary teacher snapshot

Table 1. Dataset and stream statistics used in the experiments.

Dataset	Classes	Image size	Tasks	Classes/task	Protocol
Split-CIFAR100	100	32×32	10	10	class-incremental, single pass
Split-ImageNet100	100	224×224	10	10	class-incremental, single pass

4 Experiments

4.1 Experimental Settings

Experimental protocol. We adopt the exemplar-free class-incremental setting under fixed-capacity constraints. At each task, only the current data chunk is available, without storing or replaying previous samples. Results are averaged over three random seeds and three class orders.

Datasets. We evaluate the proposed method on two widely used class-incremental benchmarks: Split-CIFAR100 and Split-ImageNet100. Both datasets contain 100 classes and are divided into 10 sequential tasks with 10 classes per task. Split-CIFAR100 uses 32×32 images, whereas Split-ImageNet100 adopts a fixed 100-class subset of ImageNet resized to 224×224 . Dataset statistics are summarized in Table 1.

Compared methods. We compare against representative exemplar-free continual-learning approaches, including regularization-based methods (EWC [16], SI [17], MAS [18]), the distillation-based method LwF [19], the prototype-based approach PASS [20], and the plasticity-oriented method PlastIL [25]. A naive fine-tuning model is also included as a lower-bound reference. Replay-based and dynamic-expansion methods are excluded because they violate the fixed-capacity exemplar-free setting considered in this work.

Table 2. Controlled temporal-chunking analysis on CIFAR-100. Lower low-PI ratio and higher eNTK effective rank indicate healthier plasticity.

Access mode	A_T (%) $\uparrow$	F_T (%) $\downarrow$	eNTK rank $\uparrow$	Low-PI ratio (%) $\downarrow$
IID access	73.84 ± 0.41	4.92 ± 0.18	87.6	5.8
Intra-distribution chunking	45.91 ± 0.73	31.48 ± 0.64	49.8	22.7
Class-incremental chunking	32.46 ± 0.68	49.36 ± 0.71	31.4	34.9

Implementation details. All methods employ a ResNet-18 backbone and are trained using SGD with momentum 0.9, weight decay 5×10^{-4} , an initial learning rate of 0.1, and a batch size of 128. Each task is trained for 50 epochs. For Split-CIFAR100, we use the standard CIFAR-style first convolution layer, while Split-ImageNet100 follows the standard ImageNet configuration. Models are initialized identically for each seed. Hyperparameters are selected using a small validation split from the current training stream and then fixed for all test runs. No samples from previous tasks are used for validation or replay.

Evaluation metrics. We employ Average Accuracy (A_T) and Average Forgetting (F_T). After training on T sequential tasks, they are defined as:

$$A_T = \frac{1}{T} \sum_{i=1}^T R_{T,i}, \quad F_T = \frac{1}{T-1} \sum_{i=1}^{T-1} \max_{t \in \{1, \dots, T-1\}} (R_{t,i} - R_{T,i}), \quad (13)$$

where $R_{t,i}$ denotes the classification accuracy on the i -th task after training on the t -th task. To connect external performance with internal plasticity, we additionally report eNTK effective rank and the ratio of neurons whose PI is below τ_{dead} .

4.2 Pathological Analysis of Temporal Chunking

We first test whether temporally localized access is associated with measurable plasticity degradation. To reduce confounding factors, we construct three diagnostic streams on CIFAR-100: IID access, intra-distribution chunking, and class-incremental chunking. IID access samples mini-batches from the full training distribution and serves as an upper-bound reference. Intra-distribution chunking preserves the global class distribution inside each chunk, while class-incremental chunking assigns disjoint classes to different chunks. This analysis is used only to verify the pathology; the proposed online method does not use the IID stream or historical samples. Table 2 reports the final predictive and diagnostic results.

The intra-distribution stream removes explicit class-level shifts, yet still causes a 27.93 point accuracy drop and a 43.2% relative reduction in eNTK effective rank compared with IID access. This supports the central diagnostic claim of the paper: temporal locality can bias the optimization trajectory toward low-rank representations. The class-incremental stream further increases the low-PI neuron ratio to 34.9%, showing that distribution shift and temporal chunking act as coupled stressors. The trend is visualized in Fig. 2 and Fig. 3.

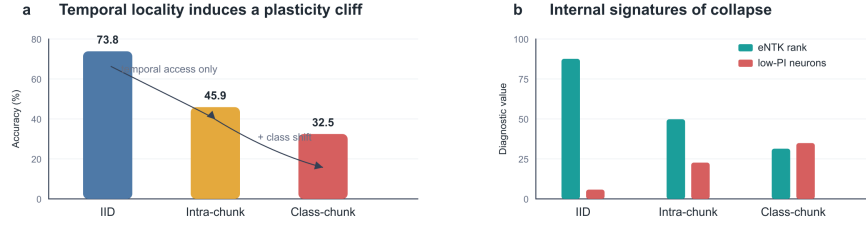


Fig. 2. Pathological analysis of temporal chunking. Pure temporal chunking already reduces accuracy and eNTK effective rank, while class-incremental chunking further accelerates representation collapse.

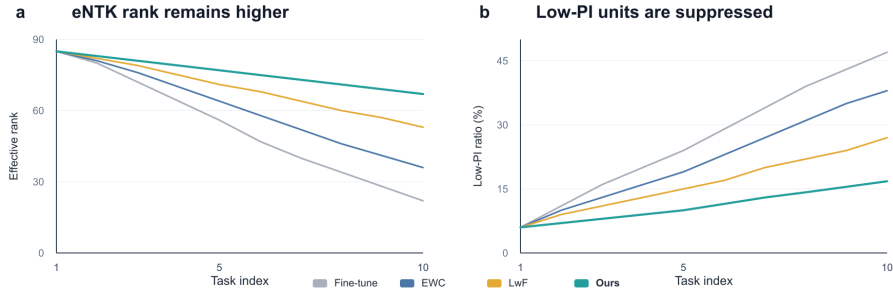


Fig. 3. Internal plasticity dynamics over the Split-CIFAR100 stream. Our method slows the decline of eNTK effective rank and suppresses the accumulation of low-PI neurons compared with fine-tuning, EWC, and LwF.

4.3 Main Results

As shown in Table 3, the proposed method achieves the best performance among all fixed-capacity exemplar-free baselines on both benchmarks. Compared with the strongest competitor PlastIL, our method improves average accuracy by 1.80% on Split-CIFAR100 and 1.57% on Split-ImageNet100, while simultaneously reducing forgetting. These results suggest that restoring dormant plastic capacity is complementary to conventional forgetting mitigation strategies. The empirical trends observed in Fig.4 are consistent with the mechanism illustrated in Fig.1(b)-(d), where plasticity diagnosis and null-space reorganization jointly preserve usable representation capacity.

4.4 Ablation Study

PI diagnosis. We first evaluate whether PI identifies genuinely low-plasticity neurons rather than merely low-magnitude parameters. As tasks accumulate, fine-tuning and EWC show a sharp increase in low-PI neurons, matching the decline of eNTK effective rank in Fig. 3. In contrast, our method keeps the final

Table 3. Main results on Split-CIFAR100 and Split-ImageNet100. We report mean $\pm$ standard deviation over three class orders and three seeds. The best result under the exemplar-free fixed-capacity setting is highlighted in bold.

Method	No exemplars	Fixed cap.	Split-CIFAR100		Split-ImageNet100	
			$A_T \uparrow$	$F_T \downarrow$	$A_T \uparrow$	$F_T \downarrow$
Fine-tune	✓	✓	19.34 $\pm$ 0.52	76.52 $\pm$ 0.63	17.85 $\pm$ 0.61	79.40 $\pm$ 0.74
EWC [16]	✓	✓	35.12 $\pm$ 0.64	43.18 $\pm$ 0.58	29.60 $\pm$ 0.66	52.85 $\pm$ 0.69
SI [17]	✓	✓	36.45 $\pm$ 0.58	41.72 $\pm$ 0.55	30.42 $\pm$ 0.64	51.36 $\pm$ 0.71
MAS [18]	✓	✓	38.06 $\pm$ 0.61	39.83 $\pm$ 0.57	32.10 $\pm$ 0.59	49.74 $\pm$ 0.65
LwF [19]	✓	✓	42.67 $\pm$ 0.47	35.40 $\pm$ 0.51	37.85 $\pm$ 0.62	43.26 $\pm$ 0.66
PASS [20]	✓	✓	52.40 $\pm$ 0.55	21.36 $\pm$ 0.46	45.20 $\pm$ 0.58	27.82 $\pm$ 0.57
PlastIL [25]	✓	✓	54.06 $\pm$ 0.61	18.24 $\pm$ 0.44	47.05 $\pm$ 0.63	25.20 $\pm$ 0.51
Ours	✓	✓	55.86$\pm$0.43	16.72$\pm$0.38	48.62$\pm$0.49	23.64$\pm$0.44

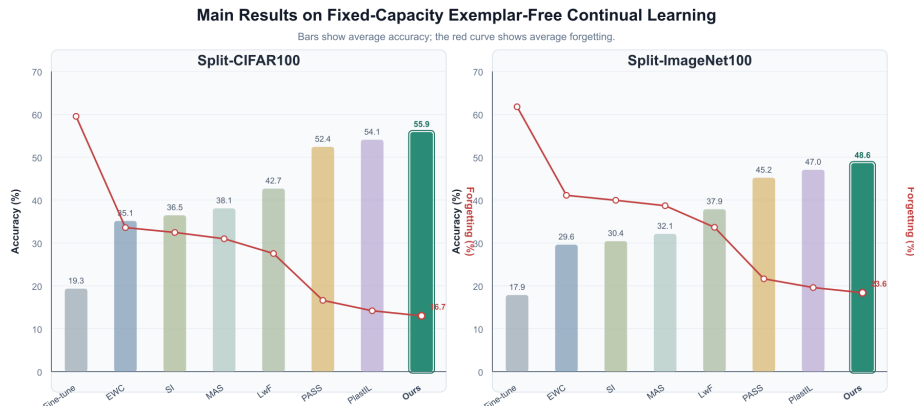


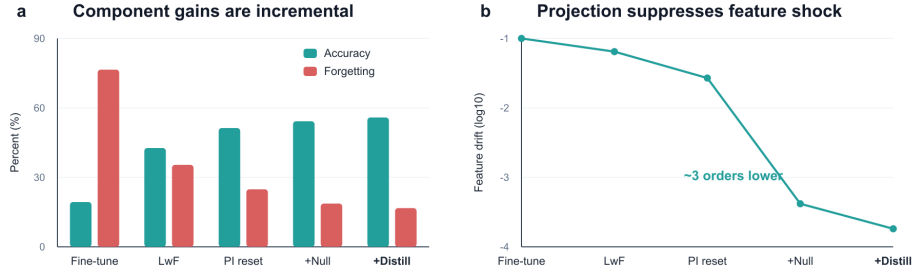
Fig. 4. Accuracy and forgetting comparison on Split-CIFAR100 and Split-ImageNet100. All compared methods avoid raw exemplar storage and backbone expansion; our method obtains a consistent but moderate advantage.

low-PI ratio below 17%, suggesting that targeted reorganization restores usable nonlinear degrees of freedom. This agrees with recent observations that plasticity loss is not fully explained by catastrophic forgetting alone [8, 9, 11].

Component ablation. Table 4 separates the contribution of the three core components on Split-CIFAR100. Direct neuron reactivation already yields substantial gains over fine-tuning, indicating that dormant units contain recoverable learning capacity. Introducing null-space projection further improves both accuracy and forgetting by suppressing interference with dominant historical feature directions. Finally, transient distillation stabilizes the reorganization phase and provides the best overall stability-plasticity trade-off. Adding null-space projection recovers stability by suppressing perturbations along historical feature directions. The transient distillation term further smooths the first updates after reorganization and yields the best accuracy-forgetting trade-off.

Table 4. Ablation study on Split-CIFAR100.

Variant	A_T (%) $\uparrow$	F_T (%) $\downarrow$	eNTK rank $\uparrow$	Feature drift $\downarrow$
Fine-tune baseline	19.34 ± 0.52	76.52 ± 0.63	26.5	1.00×10^{-1}
LwF baseline	42.67 ± 0.47	35.40 ± 0.51	42.6	6.40×10^{-2}
PI reset only	51.30 ± 0.58	24.85 ± 0.50	50.7	2.70×10^{-2}
PI + null projection	54.24 ± 0.49	18.68 ± 0.41	55.1	4.20×10^{-4}
PI + null projection + distillation	55.86 ± 0.43	16.72 ± 0.38	57.0	1.80×10^{-4}

**Fig. 5.** Ablation results. Null-space projection sharply reduces feature drift, while transient distillation improves the final stability-plasticity balance.

4.5 Resource and Sensitivity Analysis

Unlike replay-based approaches, the proposed method stores no raw exemplars and introduces no trainable parameters. The additional memory consists only of PI statistics, a compact covariance matrix, and a temporary teacher snapshot. Therefore, the overall memory overhead remains independent of the number of observed samples.

For resource accounting, all reported baselines use zero raw exemplars and no backbone growth. EWC, SI, and MAS keep parameter-importance statistics; LwF keeps a teacher snapshot; PASS stores non-raw class prototypes; PlastIL stores plasticity scores. Our method keeps PI statistics, a compact covariance matrix, and a temporary teacher snapshot. Thus the comparison isolates fixed-capacity methods that differ only in the type of non-raw auxiliary state.

We further examine two hyperparameters: the PI threshold τ_{dead} and the spectral energy ratio η used in truncated SVD. Performance is stable around $\tau_{\text{dead}} = 0.20$ and $\eta = 0.90$. Too small a threshold misses silent neurons, whereas too large a threshold reorganizes moderately useful neurons and introduces unnecessary drift. A very high η also leaves too little null space for useful updates.

5 Conclusion

This paper studies plasticity degradation in continual learning under temporally chunked access. We show that localized sequential updates are associated with

declining eNTK effective rank and an increasing proportion of low-plasticity neurons, complementing the conventional view that continual learning failure is caused only by catastrophic forgetting. To diagnose and mitigate this issue, we introduce a Plasticity Index that identifies low-plasticity units and a null-space orthogonal reorganization mechanism that reactivates these units while reducing perturbations to high-energy historical feature directions. Experiments consistently demonstrate that recovering dormant plastic capacity can serve as a practical complement to conventional forgetting mitigation strategies under strict exemplar-free and fixed-capacity constraints.

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